Modulo 5

Cloud Billing, Cost Management y AWS Support

Curso: AWS Cloud Practitioner Essentials

6 horas

Objetivos de Aprendizaje

- Explicar los modelos de precios de AWS: On-Demand, Reserved, Savings Plans, Spot
- Calcular costos de arquitecturas usando AWS Calculator y estimacion manual
- Implementar estrategias de control de costos con AWS Budgets y alertas
- Analizar costos usando AWS Cost Explorer y generar reportes
- Aplicar tagging strategies para atribucion de costos por departamento/proyecto
- Describir los diferentes planes de soporte de AWS y sus beneficios
- Utilizar AWS Trusted Advisor y Health Dashboard para optimizacion
- Implementar rightsizing y optimization recommendations
- Comparar Savings Plans vs Reserved Instances y tomar decisiones informadas

1. Modelos de Precios de AWS

On-Demand, Reserved, Savings Plans, Spot

Factores que Afectan el Costo

AWS Pricing Factors

Compute
EC2, Lambda
Tipo, Tiempo, Region

Storage
S3, EBS, Glacier
GB stored, Ops

Data Transfer
Outbound, Inter-AZ
CloudFront reduces

Network
NAT GW, ELB, VPN
Egress charged

Managed Services
RDS, Lambda, Fargate, ECS
Per-request + compute time

Support Plans
Basic (free) to Enterprise
$0 to $15,000/month

Data Transfer Rules

Data IN
FREE

Data OUT
Charged per GB

Inter-AZ Transfer
Charged

Tip: Use CloudFront to reduce data transfer costs by caching at edge locations

Reglas de Data Transfer

- **Data IN:** Gratuito
- **Data OUT:** Se cobra por GB
- **Transfer entre AZs:** Tiene costo
- **Transfer via CloudFront:** Mas barato que directo

Modelos de Precios Compute

AWS Pricing Models - Compute

On-Demand
Sin compromiso
Pago por segundo
100%
Precio base
Maxima flexibilidad

Reserved Instance
1-3 anos compromiso
Pago anticipado
28-72%
Ahorro
Menor flexibilidad

Savings Plans
Compromiso de gasto
$/hora flexibilidad
30-70%
Ahorro
Mas flexible que RI

Spot Instances
Capacidad sobrante
2 min aviso interrupcion
60-90%
Ahorro
Puede ser interrumpido

When to Use Each Model

On-Demand
Workloads impredecibles
Desarrollo y testing
Spikes no anticipadas
Primera vez en AWS

RI / SP
Baseline de produccion
Workloads predecibles
Compromiso 1-3 anos
Cost optimization

Savings Plans
Compute flexible
Mix de familias EC2
Lambda, Fargate
Gasto mensual fijo

Spot
Batch processing
ML training
Fault-tolerant
CI/CD pipelines

On-Demand Instances

Caracteristicas:

- Sin compromiso de pago
- Pago por segundo (Linux, despues del primer minuto)
- Pago por hora (Windows, otros)
- Mayor flexibilidad, mayor costo

Casos de uso: Workloads cortos, desarrollo/testing, spikes no anticipadas

Reserved Instances (RI)

Tipo	Descuento	Flexibilidad
Standard	Hasta 72%	Menos convertible
Convertible	Hasta 54%	Puede cambiar familia
Scheduled	Variable	Para windows especificos

Capacidad Reservada: Garantiza disponibilidad en AZ especifica para cargas criticas

Savings Plans

Alternativa a RIs con mas flexibilidad

- Compromiso de gasto mensual (USD/hora)
- Cubren: EC2, Lambda, Fargate

Plan	Descripcion	Ahorro
Compute SP	70% off On-Demand	Mas flexible
EC2 Instance SP	72% off On-Demand	Menos flexible

Spot Instances

Concepto: Capacidad sobrante de AWS hasta 90% descuento

Interrupcion: 2 minutos de aviso

Estrategias de uso:

- Batch processing (CI/CD builds)
- ML training (checkpoints para resumir)
- Fault-tolerant workloads
- Data analytics (Spark, Hadoop)

EC2 Fleet y Spot Fleet

EC2 Fleet:

- Gestiona múltiples instance types y AZs
- Allocation strategy: lowestPrice, diversified, capacityOptimized
- Balance between Spot y On-Demand

Spot Fleet vs Auto Scaling Group:

ASG	Spot Fleet
Scale based on metrics	Define target capacity
Can use Spot or On-Demand	Diversify across pools
Good for production	Better for fault-tolerant batch

Free Tier y Costo Control

Free Tier (6 meses):

Servicio	Beneficio
EC2	750h t2.micro Linux/Windows
S3	5GB, 20k GET, 2k PUT
RDS	750h db.t2.micro
Lambda	1M requests, 400k GB-seconds
CloudFront	1TB outbound, 10M requests

Alertas de Free Tier:

Configurar billing alert > $0 antes de límite

Lab 5.1: Estimacion, Control y Optimizacion de Costes

90 minutos

Pasos del Lab

1. Explorar Billing Dashboard
2. Crear Budget mensual con alertas
3. Configurar Cost Explorer y crear reports
4. Implementar Tagging Strategy
5. Crear Report con Tags
6. Revisar Rightsizing Recommendations
7. Configurar Billing Alerts
8. Estimar costo de arquitectura con AWS Calculator

2. AWS Cost Management Tools

Cost Explorer, Budgets, Rightsizing

Herramientas de Gestion de Costos

AWS Cost Management Tools

Cost Explorer
Visualizacion de costos
Reports personalizados
Trends y forecasting
Recommendations
Free tier: 12 months data

AWS Budgets
Alertas personalizadas
Monthly, quarterly, annual
RI/Savings Plans alerts
SNS notifications
2 budgets free, then $0.05/day

Cost Anomaly Detection
ML-powered detection
Alertas automaticas
Historial de anomalias
Account/service level
Included free

Rightsizing Recommendations
Analisis de instancias
Subutilizadas
Recommendations precisas
Preserva RI coverage
Free to use

Cost Categories
Agrupar costos logicamente
Mapear a business units
Advanced filtering
Chargeback/Showback
Included in Cost Explorer

Savings Summary
RI Coverage reports
Savings Plans utilization
Total savings tracking
Monthly trends
Included free

AWS Cost Explorer

Dashboards disponibles:

- Cost and Usage (by service, linked account)
- RI Coverage (coverage by RI vs On-Demand)
- RI Utilization (how well RIs are used)
- Savings Plans Coverage
- Savings Plans Utilization

AWS Budgets

Tipo	Descripcion	Uso
Cost	Alert cuando costo real o forecast	Control general
Usage	Alert por uso de recurso	Control especifico
RI Coverage	% instancias cubiertas por RI	RI planning
RI Utilization	% RI siendo usados	RI optimization

AWS Rightsizing Recommendations

Que es: Analisis de instancias subutilizadas con recommendations para reducir tamano

```
Instance: i-1234567890abcdef0
Current: t3.large ($70/month)
Recommended: t3.micro ($13/month)
Savings: $57/month (81%)
```

Lab 5.2: Analisis de Savings Plans vs Reserved Instances

30 minutos

Pasos del Lab

1. Analizar uso actual de EC2 con Cost Explorer
2. Escenario On-Demand (baseline)
3. Escenario Reserved Instance 1-Year
4. Escenario Compute Savings Plan
5. Escenario Reserved Instance 3-Year
6. Crear matriz de comparacion
7. Analisis de cobertura y recomendacion

3. Tagging Strategies

Organizacion, Cost Allocation, Politicas

Fundamentos de Tagging

Que es tagging? Metadatos en recursos AWS como key-value pairs

```
Key: Value
department: engineering
environment: production
project: ecommerce
cost-center: CC-12345
owner: juan@empresa.com
```

Tagging Policy y Resource Groups

- **Tagging Policy:** Reglas que requieren ciertos tags en AWS Organizations
- **Resource Groups:** Coleccion de recursos por tags para operacion en grupo
- **Tag Editor:** Central location para manage tags de todos los recursos

Cost Allocation con Tags

Ejemplo de reporte por departamento:

```
Engineering:    $2,450.00 (42%)
Marketing:      $1,890.00 (33%)
Finance:        $1,200.00 (21%)
Untagged:       $200.00 (4%)
```

4. AWS Support Plans

Basic, Developer, Business, Enterprise

AWS Support Plans Comparison

AWS Support Plans Comparison

Basic
FREE

Documentation
AWS re:Post community
Trusted Advisor (7 checks)
Personal Health Dashboard
CloudWatch (1 alarm/month)

Para: learning, personal

Developer
$29/mo

Basic +
Business hours email
Response: 12 hours
General guidance
System impaired support

Para: dev, testing

Business
$100/mo

Developer +
24/7 phone, chat, web
<1h response (production down)
Trusted Advisor full checks
TAM (with Enterprise)

Para: production

Enterprise
$15,000/mo

Business +
<15min response (critical)
Dedicated TAM
Concierge Support Team
Well-Architected Review

Para: mission-critical

Response Time SLA by Severity

AWS Trusted Advisor

Checks por categoria:

Cost Optimization	Security	Fault Tolerance
Unused EC2 instances	SG unrestricted	RDS backup check
Unattached EBS volumes	IAM access keys	Multi-AZ deployments
Low RI coverage	MFA on root	ASG health checks
Idle load balancers	S3 public access	

AWS Health Dashboard

Service Health

- Status de todos los servicios
- Historial de incidentes
- RSS feed

Personal Health Dashboard

- Alertas personalizadas
- Impacto en tus recursos
- Replacement events

5. AWS Marketplace

Soluciones de terceros listas para AWS

Introduccion a AWS Marketplace

Que es AWS Marketplace?

- Catalogo de soluciones de terceros
- Software listo para ejecutar en AWS
- SaaS, containers, AMIs, professional services

Categorias:

- Machine Learning, DevOps, Security
- Database, Analytics, Business Applications
- IoT, Services (consulting, training)

Tipos de Productos

Tipo	Descripcion	Ejemplos
SaaS	Software como servicio	Trend Micro, Splunk
AMI	Machine images	CentOS, Ubuntu Pro
Containers	ECR images	Docker containers
Consulting	Professional services	Implementation help

Uso de AWS Marketplace

Proceso:

1. Browse AWS Marketplace
2. Subscribe to product
3. Deploy (automatic or manual)
4. Usage tracking and billing

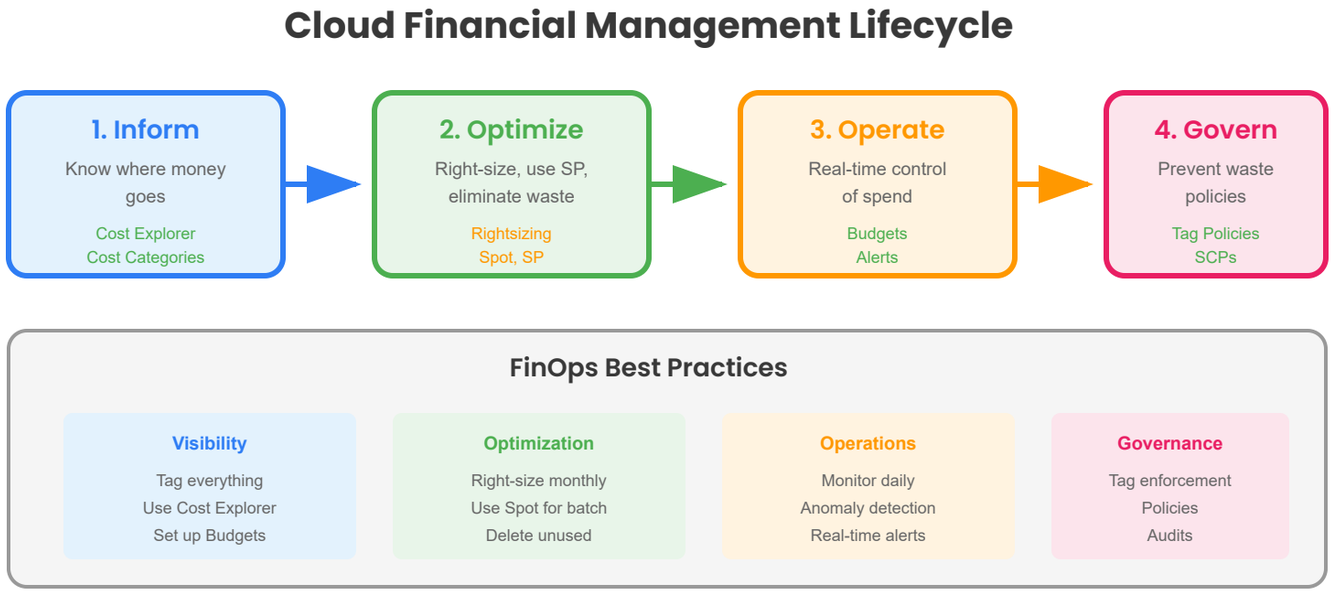
Beneficios:

- Quick deployment
- Pre-configured templates
- Compliance certifications
- Consolidated invoicing

6. Cloud Financial Management

CFM Lifecycle y FinOps

CFM Lifecycle



Continuous Optimization

Action	Impact	Esfuerzo
Right-size instances	20-40% savings	Low
Use Spot for batch	60-90% savings	Medium
Delete unused resources	5-15% savings	Low
Use S3 Intelligent-Tiering	30-40% savings	Low
Apply RIs for baseline	30-72% savings	Low

7. Estimacion de Costos - Ejercicios

AWS Pricing Calculator

AWS Pricing Calculator

Ejercicio: Web Application

```
Requirements:
- 2 EC2 t3.medium (production, 24/7)
- 1 RDS db.t3.large (Multi-AZ)
- 100GB RDS storage
- 1TB S3 storage
- 500GB outbound data transfer
```

Comparativa On-Premise vs AWS

```
On-Premise (yearly):
Hardware: $50,000 | Power/HVAC: $10,000
Staff: $150,000 | Maintenance: $5,000
Total: $215,000/year

AWS (same workload):
EC2 (t3.large x2): $2,500/year
RDS Multi-AZ: $3,200/year
Storage S3: $500/year | Data transfer: $800/year
Total: ~$7,000/year

Savings: 97% (asumiendo full utilization)
```

Resumen

Puntos Clave

- **On-Demand:** Sin compromiso, mayor costo, maxima flexibilidad
- **Reserved:** 1-3 anos compromiso, hasta 72% ahorro
- **Savings Plans:** Compromiso de gasto flexible, similar a RI
- **Spot:** Capacidad sobrante, hasta 90% descuento, interrupciones
- **Cost Explorer:** Visualizacion de costos, reports, tendencias
- **Budgets:** Alertas personalizadas, control proactivo
- **Tags:** Organizacion, atribucion de costos, politicas
- **Support Plans:** Basic (free) to Enterprise ($15k), respuesta SLA

Referencias

Recursos Adicionales

- AWS Pricing
- AWS Cost Management
- AWS Support Plans
- AWS Trusted Advisor
- AWS Pricing Calculator
- AWS Skill Builder

Gracias!

Modulo 5 Completado

Siguiente: Modulo 6 - Bases de Datos